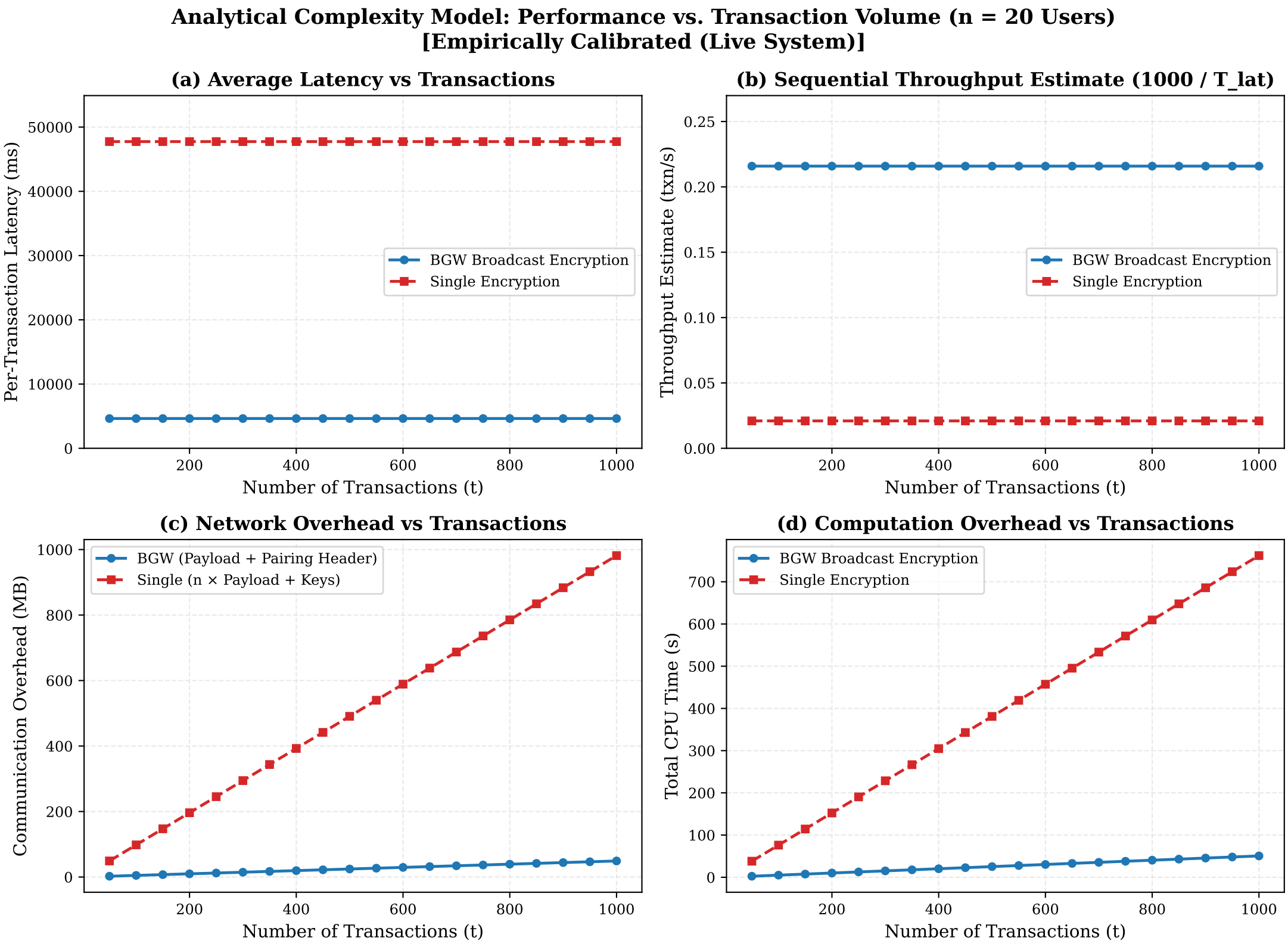
Analytical Complexity Model: Performance vs. Transaction Volume (n = 20 Users)
[Empirically Calibrated (Live System)]
(a) Average Latency vs Transactions
Per-Transaction Latency (ms)
Number of Transactions (t)
BGW Broadcast Encryption
Single Encryption
(b) Sequential Throughput Estimate (1000 / T_lat)
Throughput Estimate (txn/s)
Number of Transactions (t)
BGW Broadcast Encryption
Single Encryption
(c) Network Overhead vs Transactions
Communication Overhead (MB)
Number of Transactions (t)
BGW (Payload + Pairing Header)
Single (n × Payload + Keys)
(d) Computation Overhead vs Transactions
Total CPU Time (s)
Number of Transactions (t)
BGW Broadcast Encryption
Single Encryption